



BLUE OCEAN
ATS

BOATS FIX Drop V1.13.1

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1 Overview

The BOATS FIX Drop Copy protocol is the application layer message standard used to facilitate the electronic exchange of information related to client trades executed on the BOATS exchange. BOATS FIX Drop Copy is a text key=value formatted protocol based on the FIX Trading Community FIX 5.0 SP2 specifications. Clients can also receive order entry related events by opting in for Order Drop.

2 Transport Layer

The BOATS FIX Drop Copy protocol uses the Transmission Control Protocol (TCP) to provide reliable and ordered delivery of messages between clients and BOATS servers.

3 Session Layer

The BOATS FIX Drop Copy protocol uses the [FIXT 1.1 specification](#) session messages to establish and maintain a session.

4 Application Layer

BOATS FIX Drop Copy protocol application layer messages are based on the [Financial Information Exchange \(FIX\) Protocol version 5.0](#) and are key=value text encoded.

5 Headers and Trailers

All messages begin with a *Standard Header* and must end with a *Standard Trailer*.

5.1 Standard Header

The 'Standard Header' must contain the required fields listed and can contain any of the non-required listed fields:

Tag	Field Name	Req'd	Meaning/Values
8	BeginString	Y	"FIXT.1.1" This must be the first field in the message.
9	BodyLength	Y	The length of the FIX message. This must be the second field in the message.
35	MsgType	Y	The type of message. This must be the third field in the message.
49	SenderCompID	Y	The sender's id, assigned by BOATS at on-boarding time.
56	TargetCompID	Y	"BLUE"
115	OnBehalfOfCompID	Y	MPID
34	MsgSeqNum	Y	The message sequence number.
52	SendingTime	Y	The time of message transmission in Universal Time coordinated. See UTCTimestamp in FIX 5.0 SP2 specification.
43	PossDupFlag	N	Always required for retransmitted messages, whether prompted by the sending system or as the result of a resend request.
97	PossResend	N	Required when message may be duplicate of another message sent under a different sequence number.
122	OrigSendingTime	N	Required for message resent as a result of a ResendRequest. If data is not available set to same value as SendingTime.

5.2 Standard Trailer

The 'Standard Trailer' must contain the required fields listed and can contain any of the non-required listed fields

Tag	Field Name	Req'd	Meaning/Values
10	Checksum	Y	The message checksum. This must be the last field in the message.

6 Messages

This section defines the messages that make up the protocol. For each message, it lists the fields in the message by tag id and name, whether the field is required and acceptable values or description of the field.

6.1 Session Messages

This section defines the session level messages supported.

6.2 Heartbeat (Sent by exchange and client)

Heartbeat messages are used to confirm the status of the communication link.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 0
112	TestReqID	N	Identifier included in Test Request message to be returned in resulting Heartbeat.
	Standard Trailer	Y	

6.3 Logon

The logon message is sent by the client to establish a connection to the exchange.

A logon message must be the first message sent by the application initiating a FIX session.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = A
98	EncryptMethod	Y	Required to be 0.
108	HeartBtInt	Y	Heartbeat interval, in seconds. An interval of 0 indicates there are no heartbeats. The maximum supported interval is 90 seconds.
141	ResetSeqNumFlag	N	BOATS will reject logon if this flag is 'Y'. BOATS Market Operations can reset the inbound and outbound sequence numbers on request.
1137	DefaultApplVerID	Y	Required to be 9. ("FIX50SP2")
1408	DefaultCstmApplVerID	Y	The schema version, required as of version 1.9, is used to ensure consistency across message formats.
	Standard Trailer	Y	

Note: In version 1.9 and beyond, if the DefaultCstmApplVerID (tag 1408) is invalid, the client connection is dropped.

6.4 Logout

The logout message initiates or confirms the termination of a FIX session.

This message may be sent by the exchange or by the client.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 5
58	Text	N	Text explaining reason for logout.
	Standard Trailer	Y	

6.5 Reject

The reject message should be issued when a message is received but cannot be properly processed due to a session-level rule violation.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 3
45	RefSeqNum	Y	Reference message sequence number.
371	RefTagID	N	The tag number of the FIX field that was rejected.
372	RefMsgType	N	The MsgType of the FIX message that was rejected.
58	Text	N	Explanatory reason.
	Standard Trailer	Y	

6.6 ResendRequest

The resend request is sent by the receiving application to initiate the retransmission of messages.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 2
7	BeginSeqNo	Y	Message sequence number of first message in range to be resent.
16	EndSeqNo	Y	Message sequence number of last message in range to be resent.
	Standard Trailer	Y	

6.7 SequenceReset

The sequence reset message is used to reset the incoming sequence number on the opposing side.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 4
123	GapFillFlag	N	Indicates that this message is replacing administrative or application messages which will not be resent.
36	NewSeqNo	Y	New sequence number
	Standard Trailer	Y	

6.8 TestRequest

The test request message forces a heartbeat from the opposing application.

The opposing application responds to the Test Request with a Heartbeat containing the TestReqID.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 1
112	TestReqID	Y	Identifier included in Test Request message to be returned in resulting Heartbeat.
	Standard Trailer	Y	

7 Session Management

The client establishes a session by opening a TCP connection to the exchange Memo FIX port (server) and sending a FIX logon message with the appropriate credentials and MsgSeqNum. The SenderCompID and TargetCompID will be provisioned by the exchange and will be verified by the server at Logon. If the Logon message is accepted, the server will send a logon response. If the MsgSeqNum on the inbound logon is less than expected by the server, the logon will be rejected.

If the MsgSeqNum of the inbound Logon message is greater than expected, the server will reply with a Logon, and shall send a ResendRequest for the missing messages. The client application should gap fill any session level messages (see [FIXT 1.1 specification](#) Message Recovery), but may opt to issue sequence reset messages with the GapFillFlag set.

There are no client to server business level messages specified in this Drop Copy API specification. When the client application receives the Logon response, it should compare the MsgSeqNum to the expected server sequence number. If a gap is detected, it should send a ResendRequest message to the server. The server will resend any missing business level messages with the PossDup flag set. Instead of sending session-level messages during a backfill, the server will issue sequence reset messages with the GapFillFlag set.

When the client terminates a session it should send a Logout message and wait for the Logout response from the server before closing the connection. This procedure facilitates gap detection and processing before the connection is closed.

At the end of the trading day, the FIX inbound and outbound sequence numbers are set to 1. All inbound FIX messages shall be rejected if a FIX session has not been established. The server will send a Logout if it receives a message without PossDup flag sent and with a MsgSeqNum less than the expected sequence number. In the event of a failover to a backup port, the backup port will maintain the sequence numbers of the primary and it will have the ability to replay any messages that the primary sent.

8 Application Messages From Exchange to Client

8.1 ExecutionReport

8.1.1 Trade

This execution report is sent when a trade occurs.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange. This is guaranteed to be unique for a trading day.
11	ClOrdID	Y	Unique identifier for an Order as assigned by the client. Uniqueness must be guaranteed within a single trading day.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Uniqueness is guaranteed within a single trading day.
150	ExecType	Y	F - Trade (partial fill or fill).
39	OrdStatus	Y	1 - Partially filled 2 - Filled
55	Symbol	Y	As stated in the order.
65	SymbolSfx	N	As stated in the order.
54	Side	Y	As stated in the order.
40	OrdType	Y	As stated in the order.
32	LastQty	Y	Quantity (e.g. shares) bought/sold on this (last) fill.
31	LastPx	Y	Price of this (last) fill.
151	LeavesQty	Y	Quantity open for execution.
14	CumQty	Y	Total quantity filled on the order.
6	AvgPx	Y	The calculated average price of all fills for the executed order.
44	Price	N	As stated in the order.
38	OrderQty	Y	As stated in the order.
60	TransactTime	Y	The time at which the transaction occurred.
75	TradeDate	Y	The date that the current BOATS session ends.
880	TrdMatchID	Y	Identifier assigned to the Trade by the matching system. In case of executions on orders routed to another exchange, the TrdMatchID field will be filled on a best-effort basis.
851	LastLiquidityInd	Y	Indicator to identify whether this fill was a result of a liquidity provider providing or liquidity taker taking the liquidity. 1 - AddDisplayedLiquidity 2 - RemovedLiquidity 3 - LiquidityRoutedOut* 4 - Cross* 51 - AddHidden 52 - AddMidpoint* 53 - AddNbboImprove 54 - AddNbboJoin* 60 - RemovedOnEntry* 61- ImmediateMidpointRemoveOnEntry*

Tag	Field Name	Req'd	Meaning/Values
			62 – AddDisplayedPricImprovement* 63 – AddHiddenPricImprovement* 101 - RetailAddDisplayedLiquidity 102 - RetailRemovedLiquidity 103 - RetailLiquidityRoutedOut* 104 - RetailCross* 151 - RetailAddHidden 152 - RetailAddMidpoint* 153 - RetailAddNbboImprove 154 - RetailAddNbboJoin* 160 - RetailRemovedOnEntry 161- RetailImmediateMidpointRemoveOnEntry* 162 - RetailAddDisplayedPricImprovement* 163 - RetailAddHiddenPricImprovement* Note values 51 .. 160 are custom BOATS values.
30	LastMkt	Y	Market of execution for last fill, or an indication of the market where an order was routed. Uses ISO 10383 (MIC).
1	Account	N	Account associated to the order. Clients may opt-out of receiving this field by contacting BOATS Market Operations, otherwise it is included.
453	NoPartyIDs	Y	Number of PartyID's to follow (Always 1)
447	PartyIDSource	Y	Identifies class or source of the PartyID (448) value C - Generally accepted market participant identifier
448	PartyID	Y	MPID as stated in the order, or the default MPID for the Account if not specified in the order.
452	PartyRole	Y	Identifies the type or role of the PartyID (448) specified. 12 - Executing Trader
583	LnkID	N	As stated in the order.
1151	SecurityGroup	Y	Security Listing Tape indicator ("A", "B", "C").
528	OrderCapacity	Y	As stated in the order.
	Standard Trailer	Y	

* Not currently supported by BOATS

8.1.2 Trade Cancel/Correct

A trade correct execution report is sent to indicate that the price or quantity of a trade has been corrected. A trade cancel execution report is sent to bust a previous trade.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange. This is guaranteed to be unique for a trading day.
11	ClOrdID	Y	The ClOrdID associated with the order.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Uniqueness is guaranteed within a single trading day.
19	ExecRefID	Y	The execID of the trade been referenced.
150	ExecType	Y	G - TradeCorrect H - TradeCancel

Tag	Field Name	Req'd	Meaning/Values
39	OrdStatus	Y	1 - Partially filled 2 - Filled 4 -Canceled
55	Symbol	Y	As stated in the order.
65	SymbolSfx	N	As stated in the order.
54	Side	Y	As stated in the order.
151	LeavesQty	Y	Quantity open for execution.
14	CumQty	Y	Total quantity filled on the order.
44	Price	N	As stated on the order
40	OrdType	Y	As stated on the order
38	OrderQty	Y	As stated in the order.
31	LastPx	N	The corrected price. Required if Trade Correct.
32	LastQty	N	The corrected quantity. Required if Trade Correct.
6	AvgPx	Y	The calculated average price of all fills for the executed order.
60	TransactTime	Y	UTC timestamp since epoch with millisecond precision.
880	TrdMatchID	Y	Identifier assigned to the Trade by the matching system. In case of executions on orders routed to another exchange, the TrdMatchID field will be filled on a best-effort basis.
1	Account	N	Account associated to the order. Clients may opt-out of receiving this field BOATS Market Operations, otherwise it is included.
453	NoPartyIDs	Y	Number of PartyID's to follow (Always 1)
447	PartyIDSource	Y	Identifies class or source of the PartyID (448) value C - Generally accepted market participant identifier
448	PartyID	Y	MPID as stated in the order, or the default MPID for the Account if not specified in the order.
452	PartyRole	Y	Identifies the type or role of the PartyID (448) specified. 12 - Executing Trader
583	LnkID	N	As stated in the order.
1151	SecurityGroup	Y	Security Listing Tape indicator ("A", "B", "C").
	Standard Trailer	Y	

8.2 Order Drop Execution Reports

BOATS FIX Drop Copy by default will send the execution reports listed above. If clients opt in for Order Drop, the port will also disseminate the order entry execution reports listed below.

8.2.1 New Order Acknowledge

The New Order Acknowledge execution report is sent in response to the NewOrderSingle message when the order was accepted by the exchange. This message echoes back to the requester the accepted state of the order.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange. This is guaranteed to be unique for a trading day.
11	ClOrdID	Y	As stated in the order.

Tag	Field Name	Req'd	Meaning/Values
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Uniqueness is guaranteed within a single trading day.
150	ExecType	Y	0 - New
39	OrdStatus	Y	0 - New
55	Symbol	Y	As stated in the order.
65	SymbolSfx	N	As stated in the order.
54	Side	Y	As stated in the order.
151	LeavesQty	Y	Quantity open for further execution.
14	CumQty	Y	0
6	AvgPrice	Y	0
44	Price	N	As stated in the order.
38	OrderQty	Y	As stated in the order.
40	OrdType	Y	As stated in the order.
59	TimeInForce	Y	As stated in the order.
528	OrderCapacity	Y	As stated in the order.
18	ExecInst	Y	As stated in the order.
9416	ExtendedExecInst	N	As stated in the order.
211	PegOffsetValue	N	As stated in the order.
1094	PegPriceType	N	As stated in the order.
126	ExpireTime	N	As stated in the order.
110	MinQty	N	As stated in the order.
1138	DisplayQty	N	As stated in the order.
1084	DisplayMethod	N	As stated in the order.
1087	DisplayMinInc	N	As stated in the order.
21006	ReserveReplenishTimeType	N	As stated in the order. This is a custom BOATS tag.
114	LocateReqd	N	Must always be 'N' if present in the order.
21001	SelfTradePrevention	N	As stated in the order. This is a custom BOATS tag.
2362	StpGroupID	N	As stated in the order.
21005	RiskGroupID	N	As stated in the order. This is a custom BOATS tag.
21020	RepriceFrequencyType	N	As stated in the order. This is a custom BOATS tag.
21021	RepriceBehaviorType	N	As stated in the order. This is a custom BOATS tag.
582	CustOrderCapacity	Y	As Stated in the order. Must always be '1'.
60	TransactTime	Y	The time at which the transaction occurred. UTC timestamp since epoch with millisecond precision.
75	TradeDate	Y	The date that the current BOATS session ends.
1	Account	N	Account associated to the order. Clients may opt-out of receiving this field BOATS Market Operations, otherwise it is included.
453	NoPartyIDs	Y	Number of PartyID's to follow (Always 1)
447	PartyIDSource	Y	Identifies class or source of the PartyID (448) value C - Generally accepted market participant identifier

Tag	Field Name	Req'd	Meaning/Values
448	PartyID	Y	MPID as stated in the order, or the default MPID for the Account if not specified in the order.
452	PartyRole	Y	Identifies the type or role of the PartyID (448) specified. 3 - Client ID
583	LnkID	N	As stated in the order.
5700	LocateBroker	N	As stated on the order.
	Standard Trailer	Y	

* Not currently supported by BOATS

8.2.2 Rejected Order

The Rejected Order execution report is sent in response to a NewOrderSingle message that was rejected by the exchange.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
11	ClOrdID	Y	As stated in the order.
37	OrderID	Y	This will be the same value as ExecID.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Uniqueness is guaranteed within a single trading day.
150	ExecType	Y	8 - Rejected
39	OrdStatus	Y	8 - Rejected
55	Symbol	Y	As stated in the order.
65	SymbolSfx	N	As stated in the order.
54	Side	Y	As stated in the order.
151	LeavesQty	Y	0
14	CumQty	Y	0
6	AvgPx	Y	0
44	Price	N	As stated in the order.
38	OrderQty	Y	As stated in the order.
40	OrdType	Y	As stated in the order.
60	TransactTime	Y	UTC timestamp since epoch with millisecond precision.
75	TradeDate	Y	The date that the current BOATS session ends.
103	RejectReason	Y	Reason code for order rejection. 1 - InvalidSymbol 2 - ExchangeClosed 3 - OrderExceedsLimit 6 - DuplicateOrder 18 - InvalidPriceIncrement 19 - NoNBBOAvailable 20 - NotionalValueExceedsThreshold 22 - BlockSellShortRiskRuleViolated 23 - HardToBorrowSecurityRiskRuleViolated 27 - MaxNotionalValuePerOrderRiskRuleViolated 99 - Other BOATS specific reject codes 100 - MissingSymbol 101 - MissingLocateReqd 102 - InvalidLocateReqd 103 - MissingClOrdID

Tag	Field Name	Req'd	Meaning/Values
			104 - InvalidCLOrldId 105 - MissingSide 106 - InvalidSide 107 - MissingOrderQty 108 - InvalidOrderQty 109 - MissingOrderType 110 - InvalidOrderType 111 - MissingTimeInForce 112 - InvalidTimeInForce 113 - MissingOrderCapacity 114 - InvalidCapacity 116 - MissingLimitPrice 117 - InvalidLimitPrice 118 - MissingMaxFloor 119 - InvalidMaxFloor 120 - MissingReserveReplenishAmountType 121 - InvalidReserveReplenishAmountType 122 - MissingReserveReplenishTimeType 123 - InvalidReserveReplenishTimeType 124 - MissingRandomReplenishmentValue 125 - InvalidRandomReplenishmentValue 126 - InvalidRandomReplenishValueForReserveType 127 - MissingRepriceFrequencyType 128 - InvalidRepriceFrequencyType 129 - MissingRepriceBehaviorType 130 - InvalidRepriceBehaviorType 131 - InvalidRepriceBehaviorForRepriceFrequency 132 - MissingCustomerCapacityType 133 - InvalidCustomerCapacityType 134 - MissingExpireTime 135 - InvalidExpireTime 136 - MissingPegType 137 - InvalidPegType 138 - InvalidModifierForOrderType 139 - InvalidModifiersCombination 140 - InvalidTradingSessionForOrderType 141 - InvalidTimeInForceForOrderType 142 - InvalidModifierForPegType 143 - InvalidMinQty 144 - InvalidExecInst 145 - InvalidMpid 146 - SymbolHaltedOrPaused 147 - BlockISORiskRuleViolated 148 - BlockSessionRiskRuleViolated 149 - BlockNonTestSymbolsRiskRuleViolated 150 - MaxSharesPerOrderRiskRuleBreach 151 - PricePercentCollarRiskRuleViolated 152 - PriceValueCollarRiskRuleViolated 153 - MaxADVPercentPerOrderRiskRuleBreach 154 - DailyGrossNotionalExposureRiskRuleBreach 155 - DailyNetNotionalExposureRiskRuleBreach 156 - MaxNumDuplicateOrdersRiskRuleBreach 157 - MaxOrderRateRiskRuleBreach 158 - RestrictedSecurityRiskRuleViolated 159 - InvalidSelfTradePreventionConfiguration

Tag	Field Name	Req'd	Meaning/Values
			160 - InvalidSelfTradePreventionType 161 - InvalidRiskGroupId 162 - FirmDisabled 163 - MPIDDisabled 164 - AccountDisabled 165 - CannotTradeNonTestSymbol 166 - MissingFirm 167 - MissingAccount 168 - MissingMPID 169 - MissingRiskGroup 170 - DailyMarketOrderGrossNotionalExposureRiskRuleBreach 171 - DailyMarketOrderNetNotionalExposureRiskRuleBreach 172 - MissingDispMethodType 173 - MissingFirmRiskSetting 174 - InvalidAccountMPIDToFirm 175 - InvalidPegOffsetValue 179 - MissingSTPGroupId 180 - InvalidSTPGroupId 181 - InvalidLnkId 182 - MissingLocateBroker 183 - InvalidLocateBroker
1	Account	N	Account associated to the order. Clients may opt-out of receiving this field BOATS Market Operations, otherwise it is included.
453	NoPartyIDs	Y	Number of PartyID's to follow (Always 1)
447	PartyIDSource	Y	Identifies class or source of the PartyID (448) value C - Generally accepted market participant identifier
448	PartyID	Y	MPID as stated in the order, or the default MPID for the Account if not specified in the order.
452	PartyRole	Y	Identifies the type or role of the PartyID (448) specified. 3 - Client ID
583	LnkID	N	As stated in the order.
5700	LocateBroker	N	As stated on the order.
	Standard Trailer	Y	

8.2.3 Replaced

The Replaced execution report is sent when an order has been replaced.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange. This is guaranteed to be unique for a trading day.
11	ClOrdID	Y	As stated in the replace request.
41	OrigClOrdID	Y	As stated in the replace request.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Uniqueness is guaranteed within a single trading day.
150	ExecType	Y	5 - Replaced

Tag	Field Name	Req'd	Meaning/Values
39	OrdStatus	Y	0 - New 1 - PartiallyFilled
55	Symbol	Y	As stated in the replace request.
65	SymbolSfx	N	As stated in the replace request.
54	Side	Y	As stated in the replace request.
151	LeavesQty	Y	Quantity open for execution.
14	CumQty	Y	Total quantity filled on the order
6	AvgPx	Y	The calculated average price of all fills for the executed order.
44	Price	N	As stated in the replace request.
38	OrderQty	Y	As stated in the replace request.
40	OrdType	Y	As stated in the replace request.
59	TimeInForce	Y	As stated in the replace request.
1138	DisplayQty	N	As stated in the replace request.
114	LocateReqd	N	As stated in the replace request.
60	TransactTime	Y	The time at which the transaction occurred. UTC timestamp since epoch with millisecond precision.
75	TradeDate	Y	The date that the current BOATS session ends.
1	Account	N	Account associated to the order. Clients may opt-out of receiving this field BOATS Market Operations, otherwise it is included.
453	NoPartyIDs	Y	Number of PartyID's to follow (Always 1)
447	PartyIDSource	Y	Identifies class or source of the PartyID (448) value C - Generally accepted market participant identifier
448	PartyID	Y	MPID as stated in the order, or the default MPID for the Account if not specified in the order.
452	PartyRole	Y	Identifies the type or role of the PartyID (448) specified. 3 - Client ID
583	LnkID	N	As stated on the replace request, or the original order if not supplied on the replace request.
5700	LocateBroker	N	As stated in the replace request.
	Standard Trailer	Y	

8.2.4 Canceled

The Canceled execution report is sent when an order has been canceled. This may be because of a client requested cancel, a replace, or a cancel initiated by the exchange. Orders that are canceled by the exchange because of Time in Force instructions shall have an ExecType and OrdStatus as Expired. All other cases shall have ExecType and OrdStatus as Canceled.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange. This is guaranteed to be unique for a trading day.
11	ClOrdID	Y	As stated in the cancel request or the ClOrdID of the order if an unsolicited cancel.
41	OrigClOrdID	Y	The ClOrdID associated with the canceled order.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Uniqueness is guaranteed within a single trading day.

Tag	Field Name	Req'd	Meaning/Values
150	ExecType	Y	4 - Canceled C - Expired
39	OrdStatus	Y	4 - Canceled C - Expired
55	Symbol	Y	As stated in the cancel request.
65	SymbolSfx	N	As stated in the cancel request.
54	Side	Y	As stated in the cancel request.
151	LeavesQty	Y	0
14	CumQty	Y	Total quantity filled on the order
6	AvgPx	Y	The calculated average price of all fills for the executed order.
44	Price	N	As stated on the order.
38	OrderQty	Y	As stated on the order.
21004	CancelReason	Y	Reason code for order cancellation. This is a custom BOATS FIX tag. 0 - Other 1 - UserRequestedCancel 4 - EndOfTrading 5 - LimitUpLimitDown 6 - Halted 7 - ExchangeSupervisory 8 - OrderExpired 9 - LockOrCrossBook 10 - SelfTradePrevention 11 - InsufficientQuotes 12 - NonCompliantPrice 13 - ParticipantDisconnect 14 - OrderNotBookable 15 - TradeProtectionLimits 16 - UnableToRoute 17 - FirmDisabled 18 - MPIDDisabled 19 - AccountDisabled 20 - NotionalExposureRiskBreached 21 - InvalidCIOrdId
60	TransactTime	Y	The time at which the transaction occurred. UTC timestamp since epoch with millisecond precision.
75	TradeDate	Y	The date that the current BOATS session ends.
1	Account	N	Account associated to the order. Clients may opt-out of receiving this field BOATS Market Operations, otherwise it is included.
453	NoPartyIDs	Y	Number of PartyID's to follow (Always 1)
447	PartyIDSource	Y	Identifies class or source of the PartyID (448) value C - Generally accepted market participant identifier
448	PartyID	Y	MPID as stated in the order, or the default MPID for the Account if not specified in the order.
452	PartyRole	Y	Identifies the type or role of the PartyID (448) specified. 3 - Client ID
583	LnkID	N	As stated in the order.
	Standard Trailer	Y	

8.2.5 Order Cancel Rejected

The order cancel reject message is used by the exchange to indicate a cancel or a cancel/replace request cannot be honored. BOATS Fix Drop will send this execution report only if it can properly resolve the rejected request to recipient based on drop copy configuration.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 9
37	OrderID	Y	Unique identifier for an Order as assigned by the exchange. Uniqueness must be guaranteed within a single trading day. If the ClOrdID on the CancelRequest is unknown this field will contain "Unknown".
11	ClOrdID	Y	Unique identifier from the cancel or cancel/replace message.
41	OrigClOrdID	Y	As specified on cancel/replace request.
39	OrdStatus	Y	The status of the order after this reject was applied. 0 - New 1 - PartialFilled 2 - Filled 4 - Canceled 6 - PendingCancel 8 - Rejected A - PendingNew C - Expired E - PendingReplace
434	CxlRejResponseTo	Y	Identifies the type of request that a Cancel Reject is in response to. 1 - OrderCancelRequest 2 - OrderCancelReplaceRequest
102	CxlRejReason	Y	Code to identify reason for cancel rejection. 1 - UnknownOrder 3 - OrderAlreadyPendingCancelOrReplace 6 - DuplicateClOrdID 18 - InvalidPriceIncrement 99 - Other BOATS Custom codes 100 - MissingSymbol 101 - MissingLocate 102 - MissingClOrdID 103 - InvalidOrderQuantity 104 - InvalidSymbol 105 - InvalidLimitPrice 107 - SymbolHaltedOrPaused 108 - OrderSizeExceedsLimit 109 - ExceededMaxOrderNotionalAmt 110 - MissingOrigOrderIdentifiers 111 - AmbiguousOrigOrderIdentifiers 112 - OrigOrderSymbolNotMatchingRequestSymbol 113 - UnsupportedDisplayQuantityChange 114 - UnsupportedOrdTypeChange 115 - UnsupportedSideChange 116 - UnsupportedQuantityChange 117 - InvalidLocate 118 - ExchangeClosed 119 - BlockSessionRiskRuleViolated

Tag	Field Name	Req'd	Meaning/Values
			120 - BlockSellShortRiskRuleViolated 121 - MaxSharesPerOrderRiskRuleBreach 122 - NoNBBOAvailable 123 - MaxNotionalValuePerOrderRiskRuleBreach 124 - MaxADVPercentPerOrderRiskRuleBreach 125 - PricePercentCollarRiskRuleViolated 126 - PriceValueCollarRiskRuleViolated 127 - HardToBorrowSecurityRiskRuleViolate 128 - InvalidSide 129 - InvalidOrdType 130 - InvalidClOrdId 131 - InvalidLnkId 132 - MissingLocateBroker 133 - InvalidLocateBroker
1	Account	N	Account associated to the cancel / cancel replace request. Clients may opt-out of receiving this field BOATS Market Operations, otherwise it is included.
60	TransactTime	Y	The time at which the transaction occurred. UTC timestamp since epoch with millisecond precision.
75	TradeDate	Y	The date that the current BOATS session ends.
583	LnkID	N	As specified on the order.
5700	LocateBroker	N	As specified on the order.
	Standard Trailer	Y	

8.2.6 Restatement

The Restatement execution report is an unsolicited event sent to notify the client that an open order has been updated by the BOATS system. This message may be sent for orders configured for automatic repricing via the RepriceFrequency instruction. A restatement message will be sent if the order quantity is partially or fully cancelled because of Self Trade Prevention. This message will not be sent for Pegged Orders. In version 1.9 and beyond, clients can elect to receive an optional Restatement execution report when an order is booked and it either sets a new NBBO, or joins the existing NBBO.

NOTE: This message will be sent in the case of an order entered with the SingleReprice instruction, when the system is able to display the order at the booking price, if the order was repriced and displayed away from its booking price upon entry. In this case, the lastPx field will be the same as the previous Restatement, to indicate that the position on order book has not changed, but the message still will be sent to allow for correlation with the displayed orders on the MEMOIR Depth feed.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange. This is guaranteed to be unique for a trading day.
11	ClOrdID	Y	The ClOrdID associated with the order.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Uniqueness is guaranteed within a single trading day.
150	ExecType	Y	D - Restated
39	OrdStatus	Y	0 - New 1 - Partially filled 4 - Canceled
55	Symbol	Y	As stated in the order.

Tag	Field Name	Req'd	Meaning/Values
65	SymbolSfx	N	As stated in the order.
54	Side	Y	As stated in the order.
151	LeavesQty	Y	Quantity open for execution.
14	CumQty	Y	Total quantity filled on the order.
6	AvgPx	Y	The calculated average price of all fills for the executed order.
44	Price	N	As stated in the order.
38	OrderQty	Y	The quantity after the restatement.
31	LastPx	Y	For ExecRestatementReason (3): The updated booked price. For ExecRestatementReason (5): The match price if this order had not been prevented from executing due to STP restrictions.
32	LastShares	N	For ExecRestatementReason (5): The number of shares that would have matched if this order had not been prevented from executing due to STP restrictions.
378	ExecRestatementReason	Y	3 - RepricingOrder 5 - Partial decline of OrderQty (38) 99 - Other (When value is 99 (Other) see the ExtendedRestatementReason for more information.
60	TransactTime	Y	The time at which the transaction occurred. UTC timestamp since epoch with millisecond precision.
75	TradeDate	Y	The date that the current BOATS session ends.
1	Account	N	Account associated to the order. Clients may opt-out of receiving this field BOATS Market Operations, otherwise it is included.
453	NoPartyIDs	Y	Number of PartyID's to follow (Always 1)
447	PartyIDSource	Y	Identifies class or source of the PartyID (448) value C - Generally accepted market participant identifier
448	PartyID	Y	MPID as stated in the order, or the default MPID for the Account if not specified in the order.
452	PartyRole	Y	Identifies the type or role of the PartyID (448) specified. 3 - Client ID
21022	ExtendedRestatementReason	N	This is a custom BOATS FIX tag. 0 - None 1 - SetNBBO 2 - JoinedNBBO 3 - SelfTradeCancelNewest 4 - SelfTradeCancelOldest 5 - SelfTrade-DecrementAndCancel 6 - SelfTradeCancelBoth 7 - SelfTradeCancelSmallest If the Restatement is being sent because of Self Trade Prevention, this field contains

Tag	Field Name	Req'd	Meaning/Values
			the TYPE of Self Trade Prevention that caused the order to be reduced/canceled.
583	LnkID	N	As stated in the order.
	Standard Trailer	Y	

* Not currently supported by BOATS